

Gabrielle Browne

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Toronto, ON, Canada

Education

University of Toronto

BSc. Mathematics & Its Applications Specialist (Physical Science)

Toronto, ON

Expected Apr. 2026

George Brown College

AdvDip. Mechanical Engineering Technology – Design

Dip. Computer Numerical Control and Precision Machining

Cert. Art and Design Foundations

Toronto, ON

Aug. 2019

Publications

MorphModes: Non-rigid Registration via Adaptive Skinning Eigenmodes

Gabrielle Browne*, Mengfei Liu*, Eitan Grinspun, Otman Benckroun

under submission

Research Experience

NSERC USRA: MorphMaps

University of Toronto

May – Aug. 2025

- Single semester Undergraduate Student Research Award funded by the Natural Sciences and Engineering Research Council of Canada valued at \$8000 CAD.
- Developed a novel method for non-rigid registration by reformulating the problem into a Signed Distance Function (SDF) matching optimization problem, and using a smooth Skinning Eigenmode subspace to parameterize the problem in collaboration with the team at the Dynamic Graphics Project.
- Introduced an adaptive subspace scheme to the method to allow the resolution of localized deformations within the optimization, instead of using a competing term in the loss function for regularization in collaboration with the team at the Dynamic Graphics Project.
- Analyzed the performance of this method in comparison with existing methods.

NSERC USRA: Evaluating the Energy Efficiency of Linear Solvers in Computational Solutions of the Laplace Equation

May – Aug. 2024

University of Toronto

- Single semester Undergraduate Student Research Award funded by the Natural Sciences and Engineering Research Council of Canada valued at \$7500 CAD.
- Recorded and analyzed the accuracy, speed, and energy consumption of a selection of direct and iterative linear solvers commonly used in Computer Graphics.

Projects

CSC207 Final Project – Game Development

Aug. 2022

- Designed the rewards system for a 2D Side Scroller desktop game programmed in Java.

Mechanical Engineering Technology – Design Capstone

Apr. 2019

- Used a manual mill, drill press, and lathe to machine parts to client and group members' specifications.
- Designed a GUI in python for machinists and technologists to interface with the client's contracted device in an industrial setting.

Work Experience

**University of Toronto Information Commons
Help Desk Analyst (Contract)**

Toronto, ON
June 2023 – Dec. 2023

- Optimally utilized all channels of communication to address technical questions of staff, students, faculty, alumni, and retirees.
- Redirected questions and problems to appropriate areas of expertise outside the scope of the help desk.
- Performed identity validation interviews via video conference.
- Provided support to Assistant Help Desk Advisors regarding tickets they needed help resolving.

**University of Toronto Information Commons
Assistant Help Desk Advisor**

Toronto, ON
Jul. 2019 – June 2023

- Optimally utilized all channels of communication to address technical questions of staff, students, faculty, alumni, and retirees.
- Redirected questions and problems to appropriate areas of expertise outside the scope of the help desk.
- Performed identity validation interviews via video conference.
- Provided support for desktop computers, printers, and scanners in Robarts library, and managed basic printer maintenance.

**University of Toronto Munk School of Global Affairs & Public Policy
IT and Classroom Technology Support Analyst (Temporary)**

Toronto, ON
Feb. 2023

- Deployed computers for faculty and staff use.
- Troubleshooted Audiovisual equipment in classrooms.
- Onboarded new users via the Active Directory.
- Edited Zoom recordings of lectures.

**University of Toronto Pop Up COVID Clinic
IT Assistant**

Toronto, ON
Dec. 2021 – Feb. 2022

- Assisted with take down, security and hygiene of supplies and equipment in accordance with Ontario pandemic mandates.
- Performed data entry of client health information while maintaining confidentiality.
- Resolved technical problems with iPads and Windows machines for health professionals.

**SILVEREDGE Images Limited
On-Call Photography Assistant**

Toronto, ON
Jun. 2017 – Jul. 2019

- Prepared and organized photograph packages for delivery.
- Performed data entry of customer addresses and photo order specifications.
- Edited photographs using Adobe Lightroom and Adobe Photoshop.

**The Shoe Room at Canada's National Ballet School
Sales Associate**

Toronto, ON
Oct. 2015 – Jul. 2019

- Answered customer questions about store product and appointments both on the phone and in-person.
- Processed cash, card, and remote store transactions using Counterpoint POS software.
- Oversaw inventory and shipping processes.
- Fulfilled demi-pointe shoe and ballet slipper needs of dancers aged 1 to 80 years old.

Volunteer Experience

University of Toronto Proofs Club Organizing Committee Member

July 2025 - Present

- Designed and ran trainings for proofs club mentors.
- Structured and ran mid-semester check-in meetings for proofs club mentors.
- Hosted a proofs club kick-off event for 30 first-year undergraduate math students without prior experience writing mathematical proofs.

CS Academy

March 2025

- Organized and ran ice breakers and morning activities for a week-long March Break outreach event introducing high school students to Computer Science research.

WiGRAPH Berthouzoz Women in Research Lunch

July 2024

- Signed in attendees, directed them to their seating assignments, and acted as a runner for the luncheon.

DGP Academy

March 2024

- Organized and ran ice breakers and morning activities for a week-long March Break outreach event introducing high school students to Computer Graphics research.